

Leo Le CIS-30 Final Project Documentation

For my project I plan to create a simple personal finance tracking program (option 2) that helps users organize their monthly income, expenses, budget, and savings goals. This program will help those who frequently lose track of how much they spend or are too busy to worry about their finances. This will allow them to enter the input their financial information quickly and receive a summary of their monthly finances. I will use a simple Tkinter GUI that will contain entry boxes, buttons, and summary display area that will allow an user to input their information such as monthly budget, savings goal, income, and expenses. I will implement buttons that will allow the user to add or remove entries. I will implement this by creating empty lists for income and expenses/transactions, when the user adds income or an expense the program will validate the entry before storing it. Using a loop the program will calculate the total income and total expenses to calculate the remaining balance/budget for the month. It will also compare the expenses to the monthly budget to determine if the user is meeting the savings goal for the month. A limitation is that the user must manually input all information themselves as the program is unable to connect to bank accounts and automatically update expenses. Some improvements that could be made is adding account protection with creating accounts with passwords to view personal information or an algorithm to automatically sort expenses into categories such as entertainment, food, travel etc.

Pseudocode:

Import the graphical user interface tools

Import the custom finance tools module

Create a general Transaction class

- Store the transaction description
- Store the transaction amount
- Create a method that returns the transaction information

- Create an Income class that inherits from the Transaction class
 - Use the description and amount from the Transaction class
 - Create a method that returns the income information

- Create an Expense class that inherits from the Transaction class
 - Use the description and amount from the Transaction class
 - Store the expense category
 - Create a method that returns the expense information

- Create a BudgetTracker class
 - Set the monthly budget to zero
 - Set the savings goal to zero
 - Create an empty income list
 - Create an empty expense list

- Create a method to add an income object to the income list

- Create a method to add an expense object to the expense list

- Create a method to calculate total income
 - Set total income to zero
 - FOR each income transaction in the income list
 - Add the income amount to total income
 - END FOR
 - Return total income

- Create a method to calculate total expenses
 - Set total expenses to zero
 - FOR each expense transaction in the expense list
 - Add the expense amount to total expenses
 - END FOR
 - Return total expenses

- Create a BudgetTracker object

- Create the main Personal Finance Tracker window

Display the program title

Display an input area for the user's name

Display an input area for the monthly budget

Display an input area for the savings goal

Display an income description input area

Display an income amount input area

Display an expense description input area

Display an expense category input area

Display an expense amount input area

Display an Add Income button

Display an Add Expense button

Display a View Summary button

Display a Save Summary button

Display a Clear Entries button

Display an Exit button

Display a text area for transactions and financial results

WHEN the user selects the Add Income button

 Get the income description from the income description input area

 Get the income amount from the income amount input area

 IF the description is empty

```
    Display an error message asking for a description
ELSE
    Attempt to convert the income amount into a number

    IF the income amount is less than or equal to zero
        Display an error message asking for a positive amount
    ELSE
        Create a new Income object
        Add the Income object to the income list
        Display the income transaction in the text area
        Clear the income description and amount input areas
    END IF

    IF the income amount cannot be converted into a number
        Display an error message asking the user to enter a valid number
    END IF
END IF
```

END Add Income operation

WHEN the user selects the Add Expense button

```
    Get the expense description

    Get the expense category

    Get the expense amount

    IF the expense description or category is empty
        Display an error message asking the user to complete all expense information
    ELSE
        Attempt to convert the expense amount into a number

        IF the expense amount is less than or equal to zero
            Display an error message asking for a positive amount
        ELSE
            Create a new Expense object
            Add the Expense object to the expense list
            Display the expense transaction in the text area
            Clear the expense input areas
        END IF
    END IF
```

END IF

IF the expense amount cannot be converted into a number

Display an error message asking the user to enter a valid number

END IF

END IF

END Add Expense operation

WHEN the user selects the View Summary button

Get the user's name

Get the monthly budget

Get the savings goal

Attempt to convert the monthly budget and savings goal into numbers

IF the monthly budget or savings goal is less than zero

Display an error message asking for valid positive amounts

ELSE

Store the monthly budget in the BudgetTracker object

Store the savings goal in the BudgetTracker object

Calculate total income

Calculate total expenses

Calculate remaining balance by subtracting total expenses from total income

Calculate the remaining budget by subtracting total expenses from the monthly budget

Calculate money after savings by subtracting the savings goal from the remaining balance

IF total expenses are less than or equal to the monthly budget

Set budget status to "Within Budget"

ELSE

Set budget status to "Over Budget"

END IF

IF the remaining balance is greater than or equal to the savings goal

Set savings status to “Savings Goal Can Be Reached”

ELSE

Set savings status to “Savings Goal Cannot Be Reached”

END IF

Clear the summary display area

Display the user’s name

Display the monthly budget

Display the savings goal

Display total income

Display total expenses

Display remaining balance

Display remaining budget

Display money after savings

Display the budget status

Display the savings status

END IF

IF the monthly budget or savings goal cannot be converted into a number

Display an error message asking the user to enter valid numbers

END IF

END View Summary operation

WHEN the user selects the Save Summary button

Create the financial summary using the current user information and calculated totals

```
IF no income or expense information has been entered
    Display a message asking the user to enter financial information first
ELSE
    Attempt to open a text file for writing
    Write the user's name to the file
    Write the monthly budget and savings goal to the file
    Write all income transactions to the file
    Write all expense transactions to the file
    Write the financial totals and status messages to the file
    Close the text file
    Display a message confirming that the summary was saved

    IF the text file cannot be created or written
        Display a file error message
    END IF
END IF
```

END Save Summary operation

WHEN the user selects the Clear Entries button

```
    Clear the income description input area

    Clear the income amount input area

    Clear the expense description input area

    Clear the expense category input area

    Clear the expense amount input area

    Display a message confirming that the input areas were cleared
```

END Clear Entries operation

WHEN the user selects the Exit button

```
    Ask the user to confirm that they want to exit

    IF the user confirms
```

```
    Close the Personal Finance Tracker window  
ELSE  
    Return to the program  
END IF
```

```
END Exit operation
```

Repeat the program's button and input operations while the window remains open